

# Module 2: Requirements & DFD Analysis

IEEE 830 SRS Standard, Use Case Specifications, Data Flow Diagrams & Data Dictionaries

☰ **Module II: Software Requirements Engineering — Complete 12-Topic Syllabus Tracker**

- Topic 15:** Functional Requirements (Core Behavioral Capabilities)

**Topic 17:** User Requirements (High-Level Natural Language Statements)

**Topic 19:** The Software Requirements Specification (SRS) Document

**Topic 21:** 9 Characteristics of a Good SRS (C-U-C-C-V-F-T-P-M)

**Topic 23:** Requirements Elicitation Techniques (Interviews, JAD, Prototyping)

**Topic 25:** Requirements Validation & Formal Inspection Reviews
- Topic 16:** Non-Functional Requirements (Quality Constraints & Performance)

**Topic 18:** System Requirements (Detailed Technical Specifications)

**Topic 20:** IEEE 830 Standard Format for SRS Organization

**Topic 22:** Feasibility Study (The TELOS Analytical Framework)

**Topic 24:** Requirements Analysis & Conflict Negotiation

**Topic 26:** Requirements Management & Traceability Matrices (RTM)

## Topic 15 – 18: Functional vs. Non-Functional & User vs. System Requirements

| Requirement Class      | Formal Definition & Scope                                                                                                                  | Concrete Illustrative Example                                                                              |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| 1. Functional          | Specifies <b>WHAT</b> services, computations, and transformations the software must perform in response to specific inputs.                | "The banking system shall transfer funds between accounts and send SMS confirmation within 5 seconds."     |
| 2. Non-Functional      | Specifies <b>HOW WELL</b> the system operates, defining constraints on quality attributes: Performance, Security, Availability, Usability. | "The system shall maintain 99.99% annual availability and encrypt database records using AES-256."         |
| 3. User Requirements   | High-level natural language statements + informal diagrams intended for non-technical clients and executive sponsors.                      | "Patients should be able to book doctor consultations online."                                             |
| 4. System Requirements | Structured, unambiguous technical contracts detailing software services, inputs, outputs, exceptions, and DB schemas for developers.       | "The API endpoint `POST /api/v1/appointments` shall validate doctor availability locks via Redis mutexes." |

## Topic 19 – 21: The SRS Document & IEEE 830 Standard

### The 9 Characteristics of a High-Quality SRS (C-U-C-C-V-F-T-P-M)

1. **Correct:** Every requirement accurately reflects actual customer needs.
2. **Unambiguous:** Every requirement has exactly ONE possible interpretation by developers and testers.
3. **Complete:** All critical functional capabilities, responses to errors, and constraints are specified.
4. **Consistent:** Zero contradictions between individual requirements or interface definitions.
5. **Verifiable (Testable):** A finite, cost-effective test case can prove whether the software satisfies the requirement.
6. **Feasible:** The requirement is achievable within project budget, timeline, and technology constraints.
7. **Traceable:** Each requirement has a unique identifier linked backward to user needs and forward to code/tests.
8. **Prioritized:** Requirements are ranked by business importance (Essential, Desirable, Optional).
9. **Modifiable:** Organized with a clear table of contents and cross-references for easy updates.

## Topic 22 – 26: Requirements Engineering Process (TELOS to Traceability)

**The TELOS Feasibility Analysis Framework:** - **T — Technical:** Do we have the hardware, software tools, and developer expertise? - **E — Economic:** Is the Return on Investment (ROI) and Cost-Benefit Analysis positive? - **L — Legal:** Does the product comply with GDPR, HIPAA, copyright, and regional labor laws? - **O — Operational:** Will end-users and the organizational structure accept the system? - **S — Schedule:** Can the product be delivered before the critical market deadline?

### Requirements Traceability Matrix (RTM) Sample Architecture

| Req ID   | Requirement Description  | Source            | Design Module   | Code File            | Test Case ID  |
|----------|--------------------------|-------------------|-----------------|----------------------|---------------|
| `REQ-01` | User OTP Authentication  | Client Specs §2.1 | `AuthModule`    | `auth_service.py`    | `TC-AUTH-001` |
| `REQ-02` | Payment Gateway Checkout | Client Specs §3.4 | `PaymentEngine` | `payment_gateway.py` | `TC-PAY-014`  |

### M2 Active Recall & Exam Questions

#### Q1. Explain the IEEE 830 standard structure for a Software Requirements Specification (SRS). (8 Marks)

The IEEE 830 standard organizes an SRS into 3 primary sections:

1. **Section 1: Introduction:** Purpose, Scope, Definitions, Acronyms, References, and Document Overview.
2. **Section 2: Overall Description:** Product Perspective, Product Functions, User Classes & Characteristics, Operating Environment, Design Constraints, Assumptions & Dependencies.
3. **Section 3: Specific Requirements:** Detailed Functional Requirements, External Interface Requirements (User, Hardware, Software, Comm), Performance Requirements, Security & Safety Attributes, and Verification Criteria.